fx

| x | f(x) = $e^x$ + (1/2)x |
|---|---|
| -1 | -0,13 |
| **-0,9** | **-0,04** |
| **-0,8** | **0,05** |
| -0,7 | 0,14 |
| -0,6 | 0,25 |
| -0,5 | 0,36 |
| -0,4 | 0,47 |
| -0,3 | 0,59 |
| -0,2 | 0,72 |
| -0,1 | 0,85 |

dicotomia

| i | xa | xb | xc | f(xa) | f(xb) | f(xc) | f_xa * f_xc < 0.0 | f_xb * f_xc < 0.0 | \|xc – xc_anterior\| | precisao |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | -0.900000 | -0.800000 | -0.850000 | -0.043430 | 0.049329 | 0.002415 | 1 → xb = xc | 0 | 0.050000 | 0.000100 |
| 2 | -0.900000 | *-0.850000* | *-0.875000* | -0.043430 | 0.002415 | -0.020638 | 0 | 1 → xa = xc | 0.025000 | 0.000100 |
| 3 | *-0.875000* | -0.850000 | *-0.862500* | -0.020638 | 0.002415 | -0.009145 | 0 | 1 → xa = xc | 0.012500 | 0.000100 |
| 4 | *-0.862500* | -0.850000 | *-0.856250* | -0.009145 | 0.002415 | -0.003373 | 0 | 1 → xa = xc | 0.006250 | 0.000100 |
| 5 | *-0.856250* | -0.850000 | *-0.853125* | -0.003373 | 0.002415 | -0.000481 | 0 | 1 → xa = xc | 0.003125 | 0.000100 |
| 6 | *-0.853125* | -0.850000 | *-0.851562* | -0.000481 | 0.002415 | 0.000966 | 1 → xb = xc | 0 | 0.001563 | 0.000100 |
| 7 | -0.853125 | *-0.851562* | *-0.852344* | -0.000481 | 0.000966 | 0.000242 | 1 → xb = xc | 0 | 0.000781 | 0.000100 |
| 8 | -0.853125 | *-0.852344* | *-0.852734* | -0.000481 | 0.000242 | -0.000119 | 0 | 1 → xa = xc | 0.000391 | 0.000100 |
| 9 | *-0.852734* | -0.852344 | *-0.852539* | -0.000119 | 0.000242 | 0.000061 | 1 → xb = xc | 0 | 0.000195 | 0.000100 |
| 10 | -0.852734 | *-0.852539* | ***-0.852637*** | -0.000119 | 0.000061 | -0.000029 | 0 | 1 → xa = xc | ***0.000098*** | ***0.000100*** |
| | | | | | | | | | | |
| | | | **xc = (xa + xb)/2** | | | | | ***raiz = -0.852637*** | | |